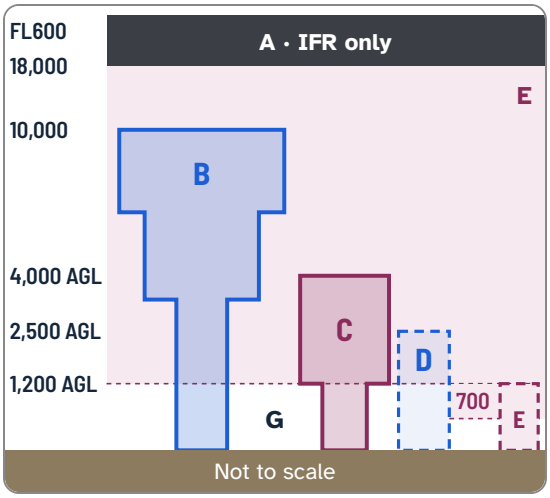
Airspace at a glance

Edition Oct 2026



Colors match the sectional chart: B solid blue, C magenta, D dashed blue, E magenta shading.

3 · 152 3 SM visibility; **1,000** above, **500** below, **2,000** to the side. Class C, D, E below 10,000 MSL, and G at night.

Special VFR: 1 SM and clear of clouds, with an ATC clearance, in a surface area below 10,000 MSL; by day unless instrument rated and IFR equipped. NO SVFR on a chart bars it for fixed-wing aircraft only.

In a surface area: no VFR beneath a ceiling lower than 1,000 ft, and ground visibility must be at least 3 SM to take off, land or enter the pattern (flight visibility if none is reported). Special VFR is the exception.

Right-of-way: converging, a balloon has it over all other categories, gliders too; an aircraft in distress, or on final or landing, has it over a balloon.

VFR weather minimums

A	18,000 MSL to FL600	—	**IFR only**
B	Any altitude	3 SM	**Clear of clouds**
C **D**	Any altitude	3 SM	**↑ 1,000 ft above ↓ 500 ft below ↔ 2,000 ft side**
E	Below 10,000 MSL	3 SM	**↑ 1,000 ft above ↓ 500 ft below ↔ 2,000 ft side**
E	At or above 10,000 MSL	5 SM	**↑ 1,000 ft above ↓ 1,000 ft below ↔ 1 SM side**

G	**Day**, 1,200 AGL or less	1 SM	**Clear of clouds**
G	**Night**, 1,200 AGL or less	3 SM	**↑ 1,000 ft above ↓ 500 ft below ↔ 2,000 ft side**
G	**Day**, above 1,200 AGL, below 10,000 MSL	1 SM	**↑ 1,000 ft above ↓ 500 ft below ↔ 2,000 ft side**
G	**Night**, above 1,200 AGL, below 10,000 MSL	3 SM	**↑ 1,000 ft above ↓ 500 ft below ↔ 2,000 ft side**
G	Above 1,200 AGL, at or above 10,000 MSL	5 SM	**↑ 1,000 ft above ↓ 1,000 ft below ↔ 1 SM side**

To enter, and who may

A	**B**	**C**	**D**	**E**	**G**
IFR, ATC clearance	**ATC clearance**	**Two-way radio contact**	**Two-way radio contact**	**None for VFR**	**None**
Instrument rating	Private; student or sport pilot with endorsement	Student	Student	Student	Student

In a balloon: Class B and C, and the airspace above them up to 10,000 MSL, need a transponder and ADS-B Out unless ATC authorizes otherwise; a balloon is not exempt there. A balloon needs neither in the 30 NM Mode C veil if it stays outside Class A, B and C and below the B or C ceiling or 10,000 MSL, whichever is lower, nor in Class E at or above 10,000 MSL (91.215(b)(3) and (b)(5), 91.225(e)). Land well short of Class D airports.

Sources: 14 CFR 91.113, 91.126–91.135, 91.155, 91.157, 61.95, 91.215, 91.225, AIM 3-2, PHAK FAA-H-8083-25C Ch. 15. Original card, not an FAA document. Study aid, not for navigation.

Commercial balloon pilot

Edition Oct 2026

What the certificate lets you do

- **Fly for hire:** act as pilot in command carrying persons or property for compensation or hire (61.133(a)(1)).
- **Teach in balloons with no CFI certificate:** give flight and ground training toward a certificate or rating, endorse for the balloon certificate and for solo, and give the training and endorsements for a flight review or recency (61.133(a)(2)(ii); 61.3(d)(3)(i)).

Logging a lesson you give: endorse the student's logbook legibly with a description of the training, the length of the lesson, your signature, your certificate number, and "certificate expiration date or recent experience end date" (61.51(h)(2)). A commercial pilot certificate has no expiration date (61.19(c)). *Instructor's practice:* name, commercial LTA certificate number, signature, date and notes on the lesson.
Pilot's note (Brian Lynch): certificates don't expire, but endorsements do, so date every one. A student's solo endorsement, for one, must come from training given within the 90 days before the flight (61.87(n)).

The airborne-heater limit (61.133(b))

Checkride in	Certificate is limited to	Limit removed by
Hot air balloon	Balloon with airborne heater	Gas balloon experience + instructor endorsement
Gas balloon	Gas balloon	Hot air experience + instructor endorsement

Medical (61.23)

Flying for hire
2nd class
Or 1st. Valid to the end of the **12th month** after the exam month, at any age

Giving flight training
None
61.23(b)(5)

Private privileges, student, or a balloon checkride
None
61.23(b)(1), (4), (10)

The medical requirement for paid balloon flying took effect May 22, 2023.

Staying current

Flight review	Since the start of the **24th calendar month** before this one: at least 1 h of flight and 1 h of ground training, including Part 91. A passed practical test counts (61.56).
Passengers	**3 takeoffs and 3 landings** in a balloon within the preceding **90 days**, as sole manipulator (61.57(a)).
Passengers at night	The same, to a full stop, between 1 h after sunset and 1 h before sunrise (61.57(b)).

Carry and show

Pilot certificate, photo ID and, when the flight needs one, your medical, on you or readily accessible. Show them on request to the FAA, the NTSB, the TSA or any law enforcement officer (61.3). Report a new permanent mailing address in writing within **30 days** (61.60).

To earn it: age 18; at least a private pilot certificate; 35 h as a pilot, including 20 h and 10 flights in balloons, 2 flights as PIC and 10 h of instruction (10+ flights) with two 1-h test-prep flights in the 2 calendar months before the test month, 2 solo flights and a controlled ascent to 3,000 ft above the launch site (61.123(a), (h); 61.129(h)).

Sources: 14 CFR 61.3, 14 CFR 61.19(c), 14 CFR 61.23, 14 CFR 61.51(h), 14 CFR 61.56, 14 CFR 61.57, 14 CFR 61.60, 14 CFR 61.87(n), 14 CFR 61.123, 14 CFR 61.129(h), 14 CFR 61.133, Final rule 87 FR 71218 (Nov. 22, 2022), FR Doc. 2022-25288. Original card, not an FAA document. Study aid, not for navigation.

The balloon itself

Edition Oct 2026

Aboard every flight

- **Airworthiness certificate**, displayed at the entrance where passengers or crew can read it (91.203(a)(1), (b)).
- **Registration certificate.** It expires 7 years after the last day of the month it was issued (91.203(a)(2); 47.40).
- **Operating limitations** in the Balloon Flight Manual or on a placard the pilot can see, and you fly within them (31.81(b); 91.9).

Required instruments (31.85)

Every balloon
Altimeter
Rate-of-climb indicator

Hot air adds
Fuel quantity gauge (each fuel cell)
Envelope temperature indicator

Captive gas adds
Compass

Inspections (91.409)

Inspection	Needed when	Interval
Annual	Every flight	Within the preceding **12 calendar months**
100-hour (an annual also satisfies it)	Carrying anyone other than crew for hire, or giving flight instruction for hire in a balloon you provide	Within the preceding **100 h** time in service. Up to 10 h over only to reach the inspection, and those hours count toward the next 100.

Who answers for what

- **Owner or operator:** keeps the balloon airworthy, including airworthiness directives (91.403(a)), and keeps records of the work, its date, and the signature and certificate number of whoever approved return to service; total time; inspection status; and each AD's number, revision date, method of compliance and next due time (91.417).
- **Pilot in command:** decides it is in condition for safe flight, and ends the flight if it becomes unairworthy (91.7).
- **Preventive maintenance a pilot may do** on a balloon the pilot owns or operates (43.3(g); a sport pilot only on a light-sport category aircraft): small envelope fabric repairs not needing load tape repair or replacement, per the manufacturer; cleaning burner pilot and main nozzles; swapping baskets and burners the type certificate data lists as interchangeable and that are designed for quick removal and installation (Part 43 App. A). Record it: what was done, the date, and your signature, certificate number and kind of certificate (43.9(a)).

Airplane rules that do not apply

91.205 instruments	Powered aircraft only
91.207 ELT	Airplanes only
91.151 fuel reserve	Airplanes and rotorcraft only. Plan propane under 91.103 and your flight manual.
91.107 belt briefing and use	A free balloon with a basket or gondola is excepted
91.215 and 91.225	Only partly: see the radio card

Sources: 14 CFR 31.81, 14 CFR 31.85, 14 CFR 43.9(a), 14 CFR 43.3(g), 14 CFR Part 43 Appendix A(c), 14 CFR 47.40, 14 CFR 91.7, 14 CFR 91.9, 14 CFR 91.107, 14 CFR 91.151, 14 CFR 91.203, 14 CFR 91.205, 14 CFR 91.207, 14 CFR 91.403, 14 CFR 91.409, 14 CFR 91.417. Original card, not an FAA document. Study aid, not for navigation.

Weather decoder

Edition Oct 2026

A METAR, group by group (example)

METAR KCRQ 211455Z 00000KT 5SM BR FEW008 BKN015 14/13 A3001 RMK AO2 SLP163 T01390128

Code	Means
METAR KCRQ 211455Z	Routine report (SPECI special), station, 21st at 1455 UTC; AUTO automated, COR corrected
00000KT	Calm. Otherwise direction from, **true**, then knots; G = gust, VRB = variable
5SM BR	Visibility 5 statute miles in mist (5/8 to under 7 SM; FG below 5/8)
FEW008 BKN015	Few at 800 ft; broken at 1,500 ft AGL, the ceiling
14/13 A3001	Temperature/dewpoint °C (M = minus); altimeter 30.01 inHg
RMK AO2 SLP163 T01390128	Automated station with a precipitation discriminator; sea-level pressure 1016.3 mb; 13.9/12.8 °C

Also in remarks: PK WND peak wind · WSHFT wind shift (FROPA: frontal passage) · PRESRR / PRESFR pressure rising / falling rapidly.

Present weather

– light, **+** heavy, none moderate; **VC** 5–10 SM away

MI shallow **PR** partial **BC** patches **DR** low drifting **BL** blowing **SH** showers **TS** thunderstorm **FZ** freezing

DZ drizzle **RA** rain **SN** snow **SG** snow grains **IC** ice crystals **PL** ice pellets **GR** hail **GS** snow pellets **UP** unknown

BR mist **FG** fog **FU** smoke **VA** volcanic ash **DU** dust **SA** sand **HZ** haze **PY** spray

PO dust/sand whirls **SQ** squalls **FC** funnel cloud (+FC tornado or waterspout) **SS** sandstorm **DS** dust storm

Sky cover

SKC or **CLR** clear (CLR: automated, nothing at or below 12,000 ft)

FEW 1/8–2/8 · **SCT** 3/8–4/8

BKN 5/8–7/8 · **OVC** 8/8

VV vertical visibility into an obscuration

Ceiling: the lowest BKN or OVC layer, or VV. Heights in hundreds of feet AGL.

TAF change groups

FM211800	From the 21st, 1800Z: a complete new forecast replaces all before it
TEMPO 2112/2115	Temporary: over 50% likely, each 1 h or less, under half the period in all
PROB30 2123/2201	30% chance of thunderstorms or precipitation
WS020/27055KT	Wind shear up to 2,000 ft AGL; wind there 270° at 55 kt

P6SM = more than 6 SM. BECMG appears in military and international TAFs, not NWS TAFs.

Winds and temps aloft (FB), example

FT 3000 6000 12000 34000

SAN 0507 2611+12 2824-01 781550

0507: 050° at 7 kt, no temp at 3,000 ft

2611+12: 260° at 11 kt, +12 °C · 2824-01: 280° at 24 kt, –1 °C

781550: direction over 50 means 100+ kt: 280° at 115 kt, –50 °C (no sign above 24,000)

9900: light and variable, under 5 kt

None within 1,500 ft of station elevation for wind, 2,500 ft for temperature.

True or magnetic?

METAR, TAF and FB winds: **true**.

ATIS, ASOS/AWOS voice, the tower, and PIREP /WV: **magnetic**.

PIREP order

Station · UA, or UUA if urgent · /OV where · /TM time UTC · /FL altitude MSL · /TP type · /SK sky · /WX weather · /TA temp · /WV wind · /TB turbulence · /IC icing · /RM remarks

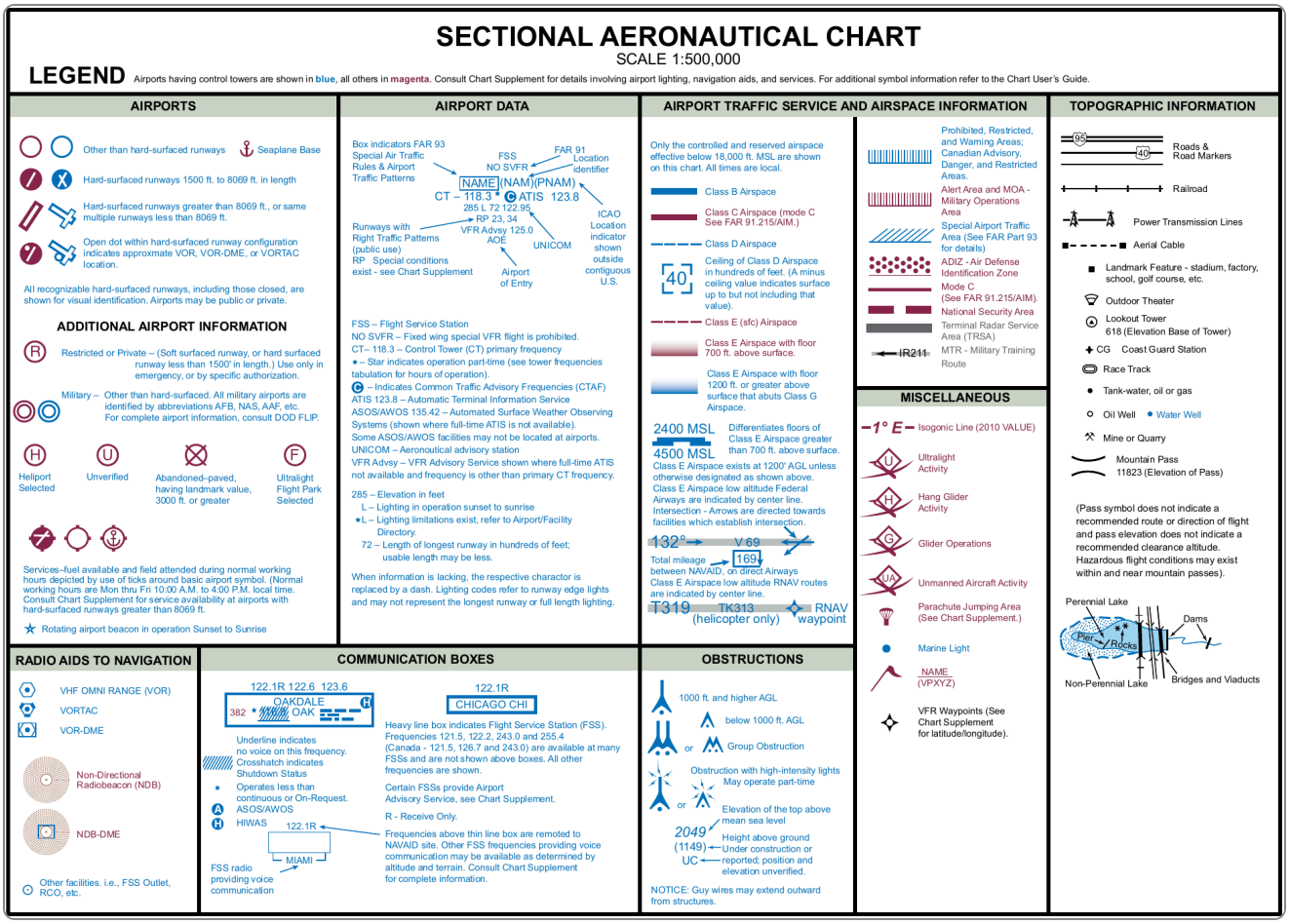
Sources: FAA-H-8083-28B Ch. 24 (Fig. 24-1, Tables 24-3, 24-4, Fig. 24-5), FAA-H-8083-28B Ch. 27 (Tables 27-1, 27-3; FB and TAF sections), AIM 7-1-10, AIM 7-1-18 (PIREP element code chart), AIM 7-1-28 and 7-1-29 (NWS TAFs exclude BECMG), AIM 4-2-10. Original card, not an FAA document. Study aid, not for navigation.

Sectional chart legend

Edition Oct 2026

The legend on your test

This is the legend page in the FAA testing supplement for the commercial test.



FAA-CT-8080-1E, Legend 1 (2018).

Changed since 2018: the Airport/Facility Directory is now the Chart Supplement, and a real chart's isogonic lines follow the current five-year magnetic variation model. The test still uses this page.

Symbols a balloon pilot reads most

A	**Obstruction** above 200 and below 1,000 ft AGL (above 299 ft in urban areas). One 1,000 ft AGL or higher gets a taller symbol.
	Power transmission line: a line broken by small tower symbols. A line with cross ticks is a railroad.
	Class C: solid magenta band.
	Class E to the surface: dashed magenta line.
	Class E from 1,200 ft AGL or higher, next to Class G: blue shading.
2049 (1149)	**Top is 2,049 ft MSL**; the number in parentheses is its height above ground (1,149 ft AGL).
	Class B: solid blue band.
	Class D: dashed blue line. A boxed number like [25] is its ceiling in hundreds of feet MSL.
	Class E from 700 ft AGL: magenta shading, fading toward the inside.
	Airport colors: blue airports have a control tower; magenta ones do not.

Sources: FAA-CT-8080-1E Legend 1, FAA Aeronautical Chart Users' Guide (Jul 2026). Original card, not an FAA document.

Study aid, not for navigation.

Operating rules

Edition Oct 2026

The pilot in command

- Final authority for the flight. In an in-flight emergency needing immediate action, deviate from any Part 91 rule as far as needed, and send a written report if the FAA asks (91.3).
- Before every flight, become familiar with all available information; for a flight not in the vicinity of an airport, that includes weather reports and forecasts, fuel requirements, and alternatives if the flight can't be completed (91.103).
- No careless or reckless operation that endangers life or property (91.13). Drop nothing that creates a hazard; dropping is allowed with reasonable precautions against injury or damage (91.15).

Right-of-way (91.113)

Aircraft in distress	Over all other traffic
Converging, different categories	**Balloon over any other category**, gliders included. Then: glider over powered aircraft; an aircraft towing or refueling over other powered aircraft; airship over the rest (91.113(d)).
Two balloons converging at about the same altitude	The one on the other's right
Head-on, or nearly so	Each pilot alters course to the right (91.113(e)) *Pilot's note (Brian Lynch): a balloon can't turn, so in a balloon you climb, descend, or brace.*
Overtaking	The one being overtaken has it; the overtaking pilot alters course to the right to pass well clear (91.113(f))
Landing	An aircraft on final approach or landing has it over aircraft in flight or on the surface. Of two approaching to land, the lower has it, but may not cut in front of or overtake one on final (91.113(g)).

Right-of-way never ends the duty to see and avoid. The pilot giving way may not pass over, under or ahead unless well clear (91.113(b)).

By custom, not regulation: the lower balloon has the right-of-way. The pilot below can't see you through the envelope. Treat it as seriously as a written rule, and before a steep climb confirm with your crew or other pilots that it is clear above you.

Balloon Flying Handbook: "There is an unwritten rule in ballooning (not regulatory) that the balloon below has the right of way (due to lack of visibility above)." (FAA-H-8083-11B, p. 6-18)

Minimum altitudes (91.119)

Except when necessary for takeoff or landing:

Anywhere
High enough for an emergency landing without undue hazard to people or property below

Congested area or open-air crowd
1,000 ft
above the highest obstacle within 2,000 ft horizontally

Elsewhere
500 ft
above the surface. Over open water or sparsely populated areas: no closer than 500 ft to any person, vessel, vehicle or structure.

No balloon exception: 91.119(d) covers only helicopters, powered parachutes and weight-shift-control aircraft.

Also

- **Alcohol and drugs (91.17):** no acting as a crewmember within **8 h** of a drink, while under the influence, at **0.04** alcohol concentration or more, or on a drug that impairs. Except in an emergency, carry no one who appears intoxicated.
- **Near other aircraft (91.111):** never close enough to create a collision hazard. Formation flight only by arrangement with each pilot, and never with passengers for hire.
- **Lights (91.209):** lighted position lights from sunset to sunrise.

Sources: 14 CFR 91.3, 14 CFR 91.13, 14 CFR 91.15, 14 CFR 91.17, 14 CFR 91.103, 14 CFR 91.111, 14 CFR 91.113, 14 CFR 91.119, 14 CFR 91.209, FAA-H-8083-11B Ch. 7 (minimum altitudes), FAA-H-8083-11B p. 6-18 (the balloon below: an unwritten rule, not regulatory). Original card, not an FAA document. Study aid, not for navigation.

NTSB Part 830

Edition Oct 2026

Is it an accident? (830.2)

An occurrence between the time anyone boards *with the intention of flight* and the time everyone has gotten off, in which anyone suffers **death or serious injury**, or the aircraft receives **substantial damage**. A fatal injury is one that causes death within **30 days**.

Serious injury

- Hospital for more than 48 h, starting within 7 days of the injury
- Any broken bone, except simple fractures of fingers, toes or nose
- Severe bleeding, or nerve, muscle or tendon damage
- Any internal organ involved
- Second- or third-degree burns, or burns over more than 5% of the body

Substantial damage

Damage or failure that harms structural strength, performance or flight characteristics and would normally need major repair or replacement of the part.

Not substantial: small punctured holes in the skin or fabric, among other listed minor damage.

Notify the NTSB immediately (830.5)

After any accident, and after any of these listed incidents even if no one is hurt:

- Flight control system malfunction or failure
- A required flight crewmember unable to perform normal duties because of injury or illness
- In-flight fire
- Aircraft collision in flight
- Damage to property other than the aircraft estimated over **$25,000** (repair, or fair market value if a total loss, whichever is less)
- An overdue aircraft believed to have been in an accident

The rest of the 830.5 list covers turbine engines, propellers, rotors, large multiengine aircraft, cockpit displays, collision-avoidance alerts and air carriers.

NTSB Response Operations Center: **844-373-9922** or 202-314-6290

What to tell them (830.6)

Aircraft type, nationality and registration · owner and operator · pilot in command · date and time · last departure point and intended landing · position from an easily defined point · number aboard, killed and seriously injured · nature of the accident, weather and damage, so far as known · any explosives, radioactive or other dangerous articles aboard.

After the call

Wreckage (830.10)	Don't move it until the NTSB takes custody, except to free injured or trapped people, protect it from further damage, or protect the public. If you must move it, sketch, note and photograph where it lay. Keep all records.
Written report (830.15)	NTSB Form 6120.1/2 within **10 days** of an accident, or after **7 days** if an overdue aircraft is still missing. For a listed incident, only if the NTSB asks. Each crewmember able to do so attaches a statement.

Sources: 49 CFR 830.2, 49 CFR 830.5, 49 CFR 830.6, 49 CFR 830.10, 49 CFR 830.15. Original card, not an FAA document. brianfliesballoons.github.io/balloon-prep

Study aid, not for navigation.

Density altitude and lift

Edition Oct 2026

Three steps

1 · Pressure altitude
PA = elevation + (29.92 − altimeter) × 1,000

2 · Standard temperature
ISA = 15 °C − 2 °C per 1,000 ft of PA

3 · Density altitude
Chart: up from OAT to the PA line, then across. Or a flight computer (E6B).

Worked example

Field elevation 1,400 ft, altimeter 29.82, OAT 30 °C.

Pressure altitude	1,400 + (29.92 − 29.82) × 1,000 = 1,400 + 100 = **1,500 ft**
Standard temperature	15 − 2 × 1.5 = **12 °C**
Density altitude	30 °C is 18° above standard. Chart: up from 30 °C to the 1,500 ft line, then across: **about 3,600 ft**

About 3,600 ft
The balloon performs as if the field were about 2,200 ft higher.

What it does to lift

- Buoyancy equals the weight of the air the envelope displaces. Heating the air inside makes it less dense, so the balloon rises when it weighs less than the air it pushes aside.
- High elevation, low pressure, high temperature and high humidity all mean thinner air and less lift at the same envelope temperature: more heat for the same load, more fuel, less margin under the envelope's limit.
- Rough rule: 1,000 cubic feet of heated air lifts about 20 lb.
- Burner performance drops about 4% per 1,000 ft.
- Maximum gross weight is in your flight manual; its performance chart reads lift from ambient temperature, envelope temperature and pressure altitude.

Worksheet

Step	Flight 1	Flight 2
Field elevation (ft)		
Altimeter setting (inHg)		
(29.92 − altimeter) × 1,000		
Pressure altitude		
ISA = 15 − 2 × (PA ÷ 1,000)		
OAT (°C)		
OAT − ISA (+ warmer, − colder)		
Density altitude (chart or E6B)		

Check your method on the FAA's own example: pressure altitude 5,048 ft at 30 °C is a density altitude of 7,855 ft (PHAK p. 8-7). The Aviation Weather Handbook computes density altitude with a flight computer (p. 8-15).

Sources: FAA-H-8083-25C Ch. 4 (standard atmosphere, pressure and density altitude), FAA-H-8083-25C p. 8-7 (worked density altitude example), FAA-H-8083-11B Ch. 2 (buoyancy), FAA-H-8083-11B Ch. 3 (performance planning), FAA-H-8083-28B p. 8-15 (density altitude: example, flight computer). Original card, not an FAA document. Study aid, not for navigation.

Radio, light signals, transponder

Edition Oct 2026

Tower light signals (AIM 4-3-13)

Signal	Aircraft in flight	Aircraft on the ground
Steady green	Cleared to land	Cleared for takeoff
Flashing green	Return for landing (steady green follows)	Cleared to taxi
Steady red	Give way to other aircraft and continue circling	Stop
Flashing red	Airport unsafe, do not land	Taxi clear of the runway in use
Flashing white	Not applicable	Return to starting point on airport
Alternating red and green	Exercise extreme caution	Exercise extreme caution

Vehicles and people: steady green go · steady red stop · flashing red clear the taxiway or runway · flashing white return to starting point.

Where a balloon needs what

Airspace	Transponder and ADS-B Out	Radio
A B C	Required unless ATC authorizes	B: ATC clearance first. C: two-way contact before entering.
Above a B or C ceiling, inside its limits, to 10,000 MSL	Required: no balloon exemption	
Mode C veil (30 NM of a listed Class B airport)	Exempt outside A, B, C and below the lower of the B or C ceiling or 10,000 MSL	
At and above 10,000 MSL, 48 states, not within 2,500 ft AGL	Exempt	
D	Not required (91.215, 91.225)	Two-way contact before entering

Exempt: balloons and aircraft certificated without an engine-driven electrical system (91.215(b)(3), (b)(5); 91.225(e)). Unequipped: request a deviation 1 h ahead. Contact is established when ATC answers with your call sign. Class D radio failure, VFR: land if basic VFR, tower in sight, cleared to land (91.129(d)(2)).

Squawk

1200 VFR · **7500** hijack
7600 lost radio · **7700** emergency

Calling

Facility, call sign, position, request: *"Palomar Tower, Balloon One Two Three Alfa Bravo, five miles east…"* Say the type and drop the N. **MAYDAY** ×3 distress, **PAN-PAN** ×3 urgency; emergency frequency **121.5**.

Alfa Bravo Charlie Delta Echo Foxtrot Golf Hotel India Juliett Kilo Lima Mike November Oscar Papa Quebec Romeo Sierra Tango Uniform Victor Whiskey Xray Yankee Zulu · 3 "tree" · 4 "fow-er" · 5 "fife" · 9 "niner"

Sources: AIM 4-3-13, AIM 4-1-20, AIM 6-2-2, AIM 6-3-1, AIM 6-3-4, AIM 6-4-2, AIM 4-2-3, AIM 4-2-4, AIM 4-2-7, AIM 3-2-4, AIM 3-2-5, 14 CFR 91.125, 14 CFR 91.129, 14 CFR 91.130, 14 CFR 91.131, 14 CFR 91.215, 14 CFR 91.225. Original card, not an FAA document. Study aid, not for navigation.

Preflight, crew and passengers

Edition Oct 2026

Before you go (91.103; PTS Area III)

- Weather reports and forecasts, PIREPs, SIGMETs and G-AIRMETs; hazards to balloon flight; a go/no-go decision.
- Fuel requirements, and alternatives if the flight can't be completed.
- A course plotted from winds aloft; airspace, obstructions and terrain; landing areas; NOTAMs.
- Operating weight, maximum load, fuel quantity and endurance; envelope temperatures, including never-exceed and maximum continuous.

Launch site (PTS V.A)

Size and surface · access and obstacles · surface wind and winds aloft · obstacles nearby · landing areas downwind · a suitable time for the conditions.

Crew briefing (PTS V.B)

- Designate a crew chief if appropriate, and give each crewmember their duties.
- Brief layout and assembly, tie-off, inflation, in flight, landing, recovery and emergency procedures.
- Agree on a common means of communication: hand signals and/or two-way radio.
- Tell them the expected direction of flight and time aloft. Check that all equipment is on board.

Layout, inspection, inflation (PTS V.C–E)

Fuel system secure, leak-free, correct pressure · cables and lines attached per the flight manual · inspect basket, envelope, suspension and handling lines, vent or deflation system, burner, fuel system, instruments · tie off as the manufacturer specifies · vent/deflation lines positioned and secured.

Passenger briefing (BFH Ch. 6, 8)

Give it before inflation and again once you decide to land:

1. **Stand where shown**: usually the rear of the basket; the front in a high-wind landing.
2. **Face the direction of travel** (or away from it, if that is what you briefed).
3. **Feet and knees together, knees bent.** Avoid the word "brace": legs stay springy.
4. **Hold on tight, in two places.**
5. **Stay in the basket** until the pilot says to get out.

No seat-belt briefing is required in a basket balloon (91.107(a)). Except in an emergency, carry no one who appears intoxicated (91.17(b)).

Pre-launch check (PTS V.F–G)

Loose items secured · wind, temperatures and obstructions reviewed · final instrument check · vent/deflation lines secured · final coordination with crew, including signals and emergency procedures · bring the balloon to equilibrium · no conflict with traffic, above and around.

Sources: 14 CFR 91.17, 14 CFR 91.103, 14 CFR 91.107, FAA-S-8081-18A Areas III and V, FAA-H-8083-11B Ch. 6 (passenger briefing), FAA-H-8083-11B Ch. 8 (passenger briefings and management). Original card, not an FAA document. Study aid, not for navigation.

Emergencies and propane

Edition Oct 2026

Fly the balloon first. Resolve the situation, then check the flight manual or checklist. In an emergency you may deviate from Part 91 as far as needed (91.3(b)). Your flight manual's procedure governs.

First moves (BFH Ch. 7–8)

Vent or deflation line lost	Only the burner controls descent. Stay low, land in the largest area you can, expect rebound and drag in wind; a poorer field may be worth it to avoid power lines.
Pilot light out	Two burners: fly on the other; try a relight using both. One burner: igniter or striker; else crack the blast valve slightly and light the stream, or use the backup system. Land as soon as practical.
Fuel leak or fire at a fitting	Shut off the fuel source. A small fire at a fitting: shut off, then snuff it with a gloved hand.
Thermal	Keep the envelope at least at its equilibrium temperature; don't vent to descend. It usually releases you higher.
High-wind landing	Lowest safe altitude to a large field, deflation line clear. Committed: brief passengers, fuel valves off, drain lines, pilot lights off. At 15 mph you cover 220 ft in the 10 s a burn takes to work.
Water landing	Passengers clear of the basket in case it inverts; hold on to it, it floats. A deflated envelope sinks; tanks float.
Power lines	Watch for them on every approach. If contact can't be avoided, follow your flight manual.

The checkride also asks about blast valve failure, fuel exhaustion and your emergency and survival gear (PTS Area X): answer from your flight manual.

Propane (BFH Ch. 2, App. A)

- Boils at −44 °F (butane 32 °F), but in the tank it is at the outside air temperature.
- Vapor is heavier than air and pools low; the odorant is ethyl mercaptan.
- Escaped liquid makes about 6,800 times its volume of combustible mixture.
- The fixed maximum liquid level gauge marks 80% full; the relief valve normally opens near 375 psi.

°F	psi
0	25
20	42
40	65
60	93
80	128
100	177

Cold mornings: add nitrogen through a regulator if the flight manual allows; vapor-feed pilot systems may not tolerate it. Cancelled, with a rise of more than 30° coming? Bleed the pressure through the fixed liquid level gauge.

Refueling: open air, never from inside the basket or a closed trailer or van · no smoking · strikers disabled · chase vehicle off · gloves. When the spit valve spits: main tank valve, then spit valve, then supply line, then bleed the lines.

Sources: 14 CFR 91.3, FAA-H-8083-11B Ch. 2 (fuel tanks, propane primer), FAA-H-8083-11B Ch. 3 (cold weather operations), FAA-H-8083-11B Ch. 7 (in-flight emergencies), FAA-H-8083-11B Ch. 8 (thermals, high-wind and water landings, refueling), FAA-H-8083-11B Appendix A, FAA-S-8081-18A Areas X and XI. Original card, not an FAA document. Study aid, not for navigation.

Checkride day

Edition Oct 2026

Before you can test (61.39)

Knowledge test passed within the **24 calendar months** before the test month · age 18 and the experience · instructor endorsement: trained within the **2 calendar months** before the month of application, ready, and weak knowledge-test areas reviewed · signed application.

Bring (PTS checklist)

For the balloon: airworthiness and registration certificates · operating limitations · maintenance records showing inspections and AD compliance · balloon flight manual.
Equipment: the PTS, a current chart, computer and plotter, flight log, current AIM.

For you: photo and signature ID · pilot certificate · medical certificate or statement (the PTS lists it; 61.23(b)(10) needs none for a balloon checkride) · FAA Form 8710-1 with your instructor's signature · knowledge test report · logbook with endorsements · Form 8060-5 if retesting · approved school graduation certificate, if any · examiner's fee.

Areas of operation (FAA-S-8081-18A)

Area	What they ask
I Fundamentals of instructing	Learning and teaching; instructor responsibilities (always) plus at least one other task
II Technical subjects	Aeromedical factors, scanning, principles of flight and false lift, Parts 1, 61, 91 and 830, airspace, the logbook entries and endorsements you will give
III Preflight preparation	Certificates, medical and inspections; weather and go/no-go; a planned maximum-duration flight with real-time weather; performance; systems
IV Preflight lesson	Teach one maneuver from Areas VII–XI as you would to a learner
V Preflight procedures	Launch site, crew briefing, layout, inspection, inflation, basket management, pre-launch check
VI Airport operations	Frequencies, phraseology, ATC clearances
VII Launches and landings	Normal and over-obstacle launches; normal and steep approaches; normal and high-wind landings. Over-obstacle launch and high-wind landing go oral if conditions don't allow.
VIII Performance maneuvers	Ascents, level flight, descents, rapid ascent and descent, contour flying, obstacle avoidance, tethering, winter and mountain flying
IX Navigation	Airspace, landmarks against the chart, position, fuel, flight duration
X Emergency operations	Malfunctions, emergency and survival gear, water landing, thermals
XI Postflight	Recovery with the landowner, deflation and pack-up, refueling and propane

Sources: 14 CFR 61.39, 14 CFR 61.23(b)(10), FAA-S-8081-18A (Nov. 2023), Applicant's Practical Test Checklist (Balloon), FAA-S-8081-18A Section 1, Areas of Operation I–XI. Original card, not an FAA document. Study aid, not for navigation.